MACHINE INTELLIGENCE (2025)

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INTRODUCTION

The objective of this practical work is to use genetic algorithms to solve different sorts of problems ranging from optimisation to generative art. The provided notebooks have been tested on Google Colab. We recommend to use Google colab. You don't need to use the GPUs in this practical work, if you run the code on Colab you can let it run a long time without getting disconnected by Google Colab if you don't use the GPUs.

PART 1

1.A MASTERMIND

Here, you will try to find the hidden code in a mastermind game using a GA and you are asked to adapt it to find a given sentence.

Task

Adapt the code to be able to find a sentence of 28 characters instead (only all caps A-Z characters and blank space) of the mastermind code.

Report

1. With your code, what would be the chromosome for the sentence "METHINKS IT IS LIKE A WEASEL"?

Submit your notebook as well

1.B TRAVELING SALESMAN PROBLEM (TSP)

Task

Based on the provided notebook, design and use a GA to the traveling salesman problem: e.g., try to find the shortest path that allows an agent to visit all the cities of a list.

The objective of this exercise is to solve the TSP problem for a set of 14 cities in Burma, officially the Republic of the Union of Myanmar. The following vectors give the GPS position of each city.

LAT = [16.47, 16.47, 20.09, 22.39, 25.23, 22.00, 20.47, 17.20, 16.30, 14.05, 16.53, 21.52, 19.41, 20.09]

LON = [96.10, 94.44, 92.54, 93.37, 97.24, 96.05, 97.02, 96.29, 97.38, 98.12, 97.38, 95.59, 97.13, 94.55]

BONUS: Consider that the distance between two cities is not the Euclidean distance. You have to consider that they are points on the surface of the Earth, which can be considered to be an oblate spheroid.

Report

1. Provide the better route you found and the shortest path in kilometers. Is it the optimal shortest path ? explain.
2. Describe your fitness function
3. Explain the way you encoded the solution, give a chromosome example.
4. Provide the configuration of the GA you finally used to find your better results: mutation, crossover, population size, type of selection, mutation, crossover used, number of generations. Describe the methodology or experiments performed in order to get your better results.
5. Provide relevant plots of your experiments and explanations.

Submit your notebook as well

PART 2

2. MONA LISA

Let's now use GA to create art, we provide a notebook to create Mona Lisa with simple shapes and your task will be to adapt it.

Task

Your task is to change this implementation in the following way:

- Instead of using polygons with 4 corners the shape must be ellipses.
- Restrict the colors to a palette of colors (For instance of 32 colors) taken from the original image.
- We don't want transparency anymore.
- Change the image to an image of your choice.

If the training time is a problem, try with fewer shapes, layers and generations. Furthemore, changing some of the GA parameters will help the model converge faster but can sometimes make it more difficult to have good results.

Report

1. The way you extracted the color palette
2. Your parameters N_SHAPES_LIST, N_INDIVIDUAL_LIST
3. Your parameters to train the genetic algorithm
4. How you define the chromosomes (what are the genes you defined and what they represent)
5. Initial image you choosed and the resulting image you obtained
6. Fitness plots

Submit your notebook as well